

Tilemachos Bolioudakis

PhD Candidate

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Education

- 2021–2026 **PhD**, *Cornell University*, Civil & Environmental Engineering, Environmental C.F.D.
2021–2024 **M.Sc. in Environmental Fluid Mechanics and Hydrology**, *Cornell University*
2019–2021 **M.Sc. in Oceanography and Management of the Marine Environment**, *NKUA*
2012–2019 **B.Sc. in Physics**, *National and Kapodistrian University of Athens (NKUA)*, Section of Environmental Physics and Meteorology

Experience

- 08/2024 – 05/2026 **System Administrator**, *Environmental C.F.D. Group*, Civil & Environmental Engineering, Cornell University
01/2023 – 05/2026 **Graduate Teaching Assistant**, *Civil & Environmental Engineering*, Cornell University, (Teaching CV attached)
08–12/2022 **Graduate Research Assistant**, *Civil & Environmental Engineering*, Cornell University
08/2021 – 08/2022 **Cornell Fellow Graduate Research Assistant**, *Civil & Environmental Engineering*, Cornell University
07–11/2020 **Volunteer**, *Youth Directorate*, Ministry of Education / Religious Affairs, Hellenic Republic

Awards

- 05/2026 **John E. Perry Teaching Assistant Prize**, *Cornell Civil & Environmental Engineering*
2022–2026 **Conference Travel Grants**, *ISSF 2022, AGU-OSM 2024, AGU-OSM 2026*
Cornell Graduate School
05–08/2023 **DeFrees Fellowship**, *Civil & Environmental Engineering*, Cornell University
2022 **Gerondelis Foundation Scholarship**, *Graduate Study Scholarship Program*, Gerondelis Foundation Inc., Lynn, MA, United States
2021 **Cornell University Fellowship**, *Academic Year 2021–2022*
2020 **Scholarship for M.Sc. Tuition Fees**, *Based on academic achievements*, National and Kapodistrian University of Athens

Publications

- 03/2026 **Published**, *Journal of Physical Oceanography*, American Meteorological Society, DOI: <https://doi.org/10.1175/JPO-D-24-0181.1>
04/2026 **Published**, *Ocean Dynamics*, Springer Nature, DOI: <https://doi.org/10.1007/s10236-026-01805-8>

Invited Talks / Conference Presentations

- 03/2026 **16th Graduate Research Symposium**, *Civil & Environmental Engineering*, Cornell University
02/2026 **American Geophysical Union**, *Ocean Sciences Meeting*, Glasgow, Scotland, UK, PS24C-1754
07/2024 **Interinstitutional Postgraduate Program NKUA-HCMR**, *Pathways to turbulence formation in high amplitude internal solitary waves of depression shoaling over gentle slopes*, Dep. of Physics, Athens, Greece, Announcement link

- 05/2024 **Fluid Dynamics of Sustainability and the Environment**, *Summer School*, Ecole Polytechnique, Paris, France, <http://fdse.org>
Poster
- 02/2024 **American Geophysical Union**, *Ocean Sciences Meeting*, New Orleans, Louisiana, USA, <https://agu.confex.com/agu/OSM24/meetingapp.cgi/Person/1469507>
Poster
- 11/2023 **American Physical Society**, *76th Annual Meeting of the Division of Fluid Dynamics*, Washington, D.C., USA, <https://meetings.aps.org/Meeting/DFD23/Session/T44>
Presentation
- 11/2022 **American Physical Society**, *75th Annual Meeting of the Division of Fluid Dynamics*, Indianapolis, Indiana, USA, <https://meetings.aps.org/Meeting/DFD22/Session/L27.2>
Presentation
- 08/2022 **9th International Symposium on Stratified Flows**, *Cambridge, UK*, <https://www.damtp.cam.ac.uk/events/issf2022/>
Poster

Approved High-Performance-Computing Proposals

- 09/2024 **Maximize Project**, *NSF supported ACCESS Allocations, USA*, Role: Co-PI, Amount: 6.5M Service Units
- 03/2021 **Preparation/Development Projects**, *GRNET HPC System A.R.I.S., Greece*, Role: Co-PI, Amount: 200K Service Units

Additional Training

- 05/2024 **Fluid Dynamics of Sustainability and the Environment**, *Summer School*, Ecole Polytechnique, Paris, France, <http://fdse.org>
- 12/2020 **GRNET PRACE Training Center**, *Efficient Use of HPC Systems*
- 11/2020 **GRNET PRACE Training Center**, *Introduction to Parallel Programming*
- 10/2018 **Hellenic Physics Union Summer School**, *Aegina, Greece*
- 07/2018 **25th summer school-conference. Dynamical systems and complexity**, *NCSR Demokritos, Athens, Greece*

Languages

- | | | |
|---------|--|---------|
| English | British Council IELTS (grade 7.0 of 9) | 05/2019 |
| Greek | Native Language | |

Computer Skills

- | | |
|-------------|--|
| Experience | Linux/Unix environments, High-Performance Computing |
| Data | Synology NAS servers, Globus data transfers |
| Software | MATLAB, TecPlot, VisIt, $\text{\LaTeX}$, Microsoft Office, ArcGIS |
| Programming | Fortran, Python, R |

Teaching CV

Teaching Awards and Recognition

- Recipient, 2026 John E. Perry Teaching Assistant Prize, Cornell Engineering
- Nominee, 2026 Cornelia Ye Outstanding Teaching Assistant Award, Center for Teaching Innovation, Cornell University

Teaching Assistant Evaluations

Average student evaluation scores (out of 5.0) during PhD studies

- Spring 2023: 3.69/5.00
- Fall 2023: 4.34/5.00
- Spring 2024: 4.33/5.00
- Fall 2024: 4.80/5.00
- Spring 2025: 4.85/5.00
- Fall 2025: N/A (Graduate-level course)
- Spring 2026: 4.55/5.00

Teaching Experience and Responsibilities

CEE (ENGRD) 3200 Engineering Computation (Undergraduate)

Teaching Assistant: Spring 2023, 2024, 2025, 2026 (ongoing)

- Led weekly recitations and review sessions supporting homework and project development.
- Held regular and exam-focused office hours for group and one-on-one guidance.
- Designed and graded problem sets, projects, and exams integrating course material with engineering applications.
- Developed computational examples connecting theory with numerical implementation.
- Responded to student questions and provided clarifications through Piazza.

CEE 3310 Fluid Mechanics (Undergraduate)

Teaching Assistant: Fall 2023, 2024

- Supported lecture discussions through board-based problem solving.
- Assisted in exam preparation and laboratory experiment design.
- Led laboratory sessions in the DeFrees Teaching Laboratory flumes and supervised junior TAs.
- Graded assignments and exams with detailed written feedback.
- Provided online clarifications through Canvas.

CEE 6000 Numerical Methods for Engineers (Graduate)

Teaching Assistant: Fall 2025

- Led advanced office hours on numerical modeling and stability analysis.
- Assisted in assignment design and grading.

Private Tutor (High School Level), Athens, Greece

During B.Sc. and M.Sc. studies

- Tutored students in Physics, Mathematics, and Chemistry, including preparation for final and university entrance examinations.